# Hypothesis Testing I

Unit - 4
Session - 7

# Frame the Hypothesis

## Questions:

1. **Retail Sales:** The average transaction time for customers at a convenience store is known to be 75 seconds. Following the installation of a new self-checkout system, the store manager wants to know if the average transaction time has been **reduced**.

2. **Educational Standard:** Historically, the mean score for high school students on a national history exam has been 500 points. A new curriculum is introduced, and educators want to see if the mean score is **higher** than 500 points.

3. **Public Health:** Government data indicates that the average lifespan of a certain brand of common light bulb is 8,000 hours. A major competitor claims that the average lifespan of their own light bulb is **less than** 8,000 hours.

4. **Traffic Safety:** The average speed on a busy highway is currently 65 mph. After police increased speed monitoring, authorities want to determine if the average speed on the highway has **changed**

## Answer:

1. **H0 ≥ 75 seconds**
   **H1 < 75 seconds**

   **Left-Tailed**

2. **H0 ≤ 500 points**
   **H1 > 500 points**

   **Right Tailed**

3. **H0 ≥ 8000 hours**
   **H1 < 8000 hours**

   **Left-Tailed**

4. **H0 = 65 mph**
   **H1 ≠ 65 mph**

   **Two-Tailed**

# Unit 4: Hypothesis Testing

- Test of Mean - One sample Z Test - Test of Mean
- One sample t-Test - Test of Mean
- 2 samples - Independent and Dependent - 2 samples
- Test of proportion - One sample (Z proportion test)
- Two samples (Z proportion test)
- Case Study - Categorical vs Continuous
- More than 2 samples

# Agenda

- Test of Mean
  - One sample (Z & T)
  - Independent and Dependent - 2 sample
  - Three or more samples (Anova) (details later)

- Test of proportion
  - One sample (Z proportion test)
  - 2 sample (Z proportion test)
  - Three or more samples (Chi Square) (details later)

# One Sample – Z test

# One-Sample test - Z test for Mean (σ known)

Z test for Mean (σ known)

$$Z_{STAT} = \frac{\overline{X} - \mu}{\sigma / \sqrt{(n)}}$$

- where X- is the mean of the sample μ is the mean of the population
  σ is the standard deviation of the population n is the number of observations

# Decision rule

| | $H_1$ | Based on critical region | Based on p-value | Based on confidence interval |
|---|---|---|---|---|
| For two tailed test | $\mu \neq \mu_0$ | Reject $H_0$ if $\lvert Z \rvert > Z_{\alpha/2}$ | Reject $H_0$ if p-value is less than or equal to the level of significance | Reject $H_0$ if $\mu_0$ does not lie in the confidence interval |
| For left tailed test | $\mu < \mu_0$ | Reject $H_0$ if $Z < -Z_\alpha$ | | |
| For right tailed test | $\mu > \mu_0$ | Reject $H_0$ if $Z > Z_\alpha$ | | |

# One-Sample test

A Pizza restaurant claims that the average delivery time of pizza is 20 minutes with the Standard deviation of 12 minutes. A sample of 30 pizza delivery times is given below:

38, 38, 10, 10, 10, 10, 10, 38, 10, 38, 10, 10, 10, 10, 10, 38, 10, 38, 10, 38, 38, 38, 10, 38, 38, 38, 38, 10,10, 38

Test whether there is a significant deviation between your sample mean and the population mean?

# One-Sample test - Z test for Mean (σ known)

A Pizza restaurant claims that the average delivery time of pizza is 20 minutes with the Standard deviation of 12 minutes. A sample of 30 pizza delivery times is given below:

38, 38, 10, 10, 10, 10, 10, 38, 10, 38, 10, 10, 10, 10, 10, 38, 10, 38, 10, 38, 38, 38, 10, 38, 38, 38, 38, 10, 10, 38

Test whether there is a significant deviation between your sample mean and the population mean?

**Solution**

We observe from the sample, **sample mean** = 692 / 30 = 23.07

**n** = 30

**μ** = 20 and **σ** = 12

# One-Sample test - Z test for Mean (σ known)

**Step 1:** State the null and alternative hypothesis

**$H_o$: μ = 20 minutes**
**$H_a$: μ ≠ 20 minutes**

**Step 2:** Choose the level of significance, α to be 5% and sample size is 30

**Step 3:** Choose the appropriate test statistic. Since σ is known, you use the normal distribution and ZSTAT as test statistic.

**Step 4:** Determine the critical value that divides the rejection and non-rejection regions.

Critical values are selected so that the rejection region contains a total area of α when $H_o$ is true and the non-rejection region contains a total area of 1 - α when $H_o$ is true. Since α = 0.05, the critical value of ZSTAT is **1.96**.

# One-Sample test - Z test for Mean (σ known)

## Critical value

**Step 5**: Collect the data, organize the results and compute the value of the test statistic.

$$Z_{STAT} = \frac{\bar{X} - \mu}{\sigma / \sqrt{(n)}}$$

Z = (23.07 - 20) / (12 / sqrt(30)) = 1.40

**Step 6**: State the statistical decision and the managerial conclusion.

- $Z_{STAT}$ is in the region of non-rejection because $Z_{STAT} < Z_{CRIT}$ ($Z_{CRIT}$ =1.96) where $Z_{STAT}$ = 1.40
- Because the test statistic falls in the non-rejection region, we fail to reject the null hypothesis.
- The managerial conclusion is that insufficient evidence exists to prove that pizza delivery time is different from the claimed delivery time of 20 minutes. No Corrective action on delivery process is required.

# One-Sample test - Z test for Mean (σ known)

## P value approach

**Step 5**: Collect the data, organize the results and compute the p value of the test statistic.

$Z = (23.07 - 20) / (12 / \text{sqrt}(30)) = 1.40$

$P(X > Z = 1.40) = 1 – 0.9192 = 0.0808$

**Step 6**: State the statistical decision and the managerial conclusion.

- Since P-value (0.0808) > 0.05 (our level of significance), we fail to reject null hypothesis.
- The managerial conclusion is that insufficient evidence exists to prove that pizza delivery time is different from the claimed delivery time of 20 minutes. No Corrective action on delivery process is required.

# Summary

- The Z test can be used if the sample data is drawn from a normally distributed population
- The normality of the dataset can be checked using Shapiro-Wilk test
- Two tailed as well as one tailed (left/ right) Z test can be performed
- The test statistic for the Z test follows normal distribution under the null hypothesis
- The test statistic, p-value and confidence interval can be used to analyze the significance of the test

# Shapiro wilk test

## Problem:

A small bakery claims cookies take **12 minutes** to bake. Recorded baking times of 5 cookies:

[11,12,12,13,14]

Check if the data is **normally distributed** using a simplified Shapiro-Wilk approach.

Check:

1. Are the baking times roughly normal? → Shapiro-Wilk

2. Is the average baking time significantly different from 12? → z-test

## Soln:

**Step 1: Order the data**

Arrange in ascending order:

x=[11,12,12,13,14]

**Step 2: Compute sample mean**

$\bar{x}$=11+12+12+13+145=62/5=12.4

**Step3:** The **Shapiro-Wilk test formula** is:

$$W = \frac{(\sum a_i x_{(i)})^2}{\sum (x_i - \bar{x})^2}$$

For **n = 5**, approximate constants from the table:

a=[0.6646,0.2413]

- Pair smallest with largest, 2nd smallest with 2nd largest. Middle value ignored.

*Table

numerator=(a1(x5−x1)+a2(x4−x2))^2

= (0.6646*(14-11)+0.2413*(13-12))^2

= 4.995

∑(xi−x¯)^2=(11−12.4)^2+(12−12.4)^2+(12−12.4)^2+(13−12.4)^2+(14−12.4)^2

=(−1.4)^2+(−0.4)^2+(−0.4)^2+0.6^2+1.6^2

=1.96+0.16+0.16+0.36+2.56

=5.2= 1.96 + 0.16 + 0.16 + 0.36 + 2.56 = 5.2

W=Numerator^2/Denominator

=4.995/5.2≈0.9606

**Step-4:**

- For n = 5, α = 0.05 → critical W ≈ 0.762

W=0.9606>0.762

**Conclusion:** Fail to reject $H_0$ → data **is roughly normal**.

# Shapiro Wilk Test - Question

A researcher collects the following 8 measurements of reaction time (in milliseconds):
520, 560, 480, 580, 510, 550, 490, 540

Check if the data is normally distributed using a simplified Shapiro-Wilk approach.

| i | yi | yi−y¯ | (yi−y¯)2 |
|---|---|---|---|
| 1 | 480 | -50 | 2500 |
| 2 | 490 | -40 | 1600 |
| 3 | 510 | -20 | 400 |
| 4 | 520 | -10 | 100 |
| 5 | 540 | 10 | 100 |
| 6 | 550 | 20 | 400 |
| 7 | 560 | 30 | 900 |
| 8 | 580 | 50 | 2500 |
| **Sum** | **4230** | **0** | **8500** |

Mean(y)=530

$$b = \sum_{i=1}^{m} a_i(y_{n-i+1} - y_i) = a_1(y_8 - y_1) + a_2(y_7 - y_2) + a_3(y_6 - y_3) + a_4(y_5 - y_4)$$

| $i$ | $y_i$ | $y_{n-i+1}$ ($y_{9-i}$) | $y_{n-i+1} - y_i$ | $a_i$ (from table, $n = 8$) | $a_i(y_{n-i+1} - y_i)$ |
|---|---|---|---|---|---|
| 1 | 480 | $y_8 = 580$ | 100 | 0.6052 | 60.52 |
| 2 | 490 | $y_7 = 560$ | 70 | 0.3164 | 22.148 |
| 3 | 510 | $y_6 = 550$ | 40 | 0.1976 | 7.904 |
| 4 | 520 | $y_5 = 540$ | 20 | 0.1224 | 2.448 |
| | | | | | **Sum ($b$) = 93.02** |

$$W = \frac{(\sum a_i(y_{n-i+1} - y_i))^2}{\sum(y_i - \bar{y})^2} = \frac{b^2}{D}$$

$$W = \frac{(93.02)^2}{8500} = \frac{8652.7204}{8500} \approx \mathbf{1.018}$$

From the table:

$$\mathbf{W}_{\text{critical}} \text{ (for } \mathbf{n = 8}, \alpha = \mathbf{0.05}) = \mathbf{0.818}$$

**Conclusion:**

- **Calculated W:** $1.018$
- **Critical W:** $0.818$

The decision rule is to **reject** $H_0$ **if** $W_{\text{calculated}} < W_{\text{critical}}$.

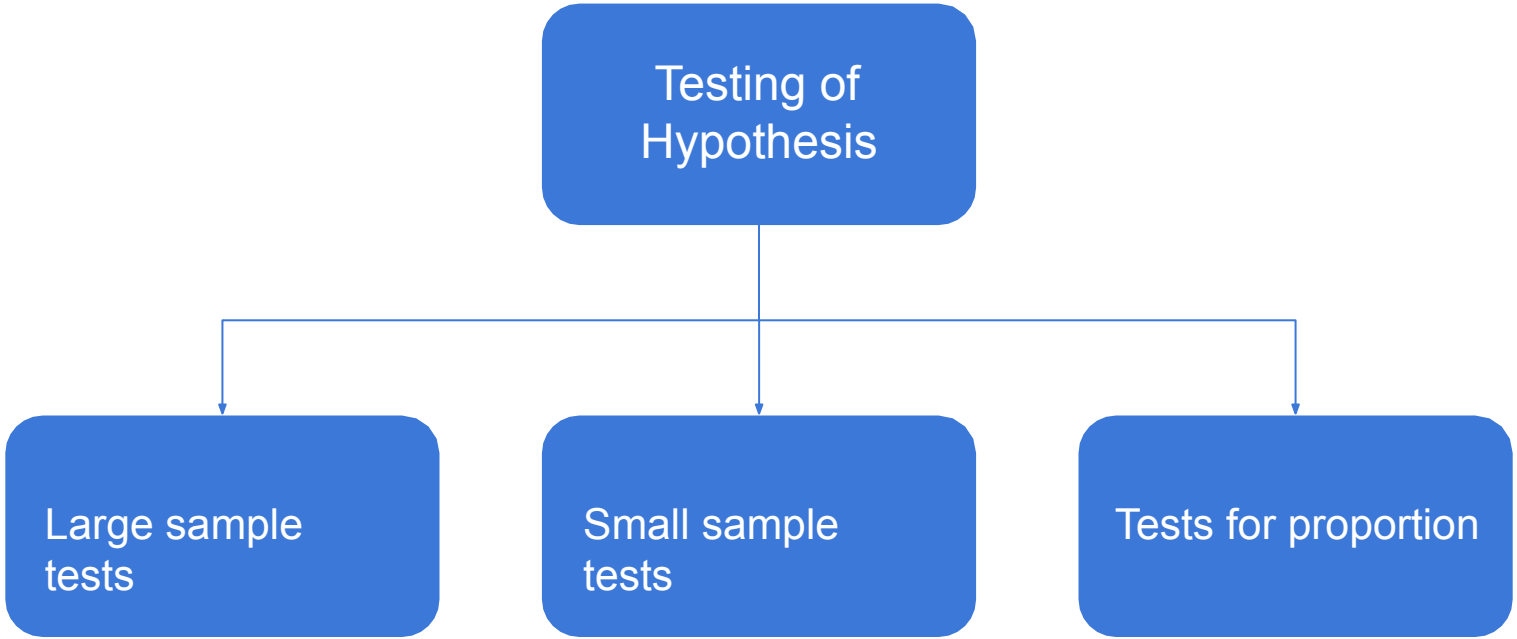
Since $1.018$ is **not less than** $0.818$ ($1.018 \not< 0.818$), we **do not reject the null hypothesis**

# One Sample – t test

# Last topic

- The unknown population parameters are estimated by from sample
- The unknown population parameters are replaced by the sample estimates
- Test the hypothesis for population mean when the population variance is known

# In this session…

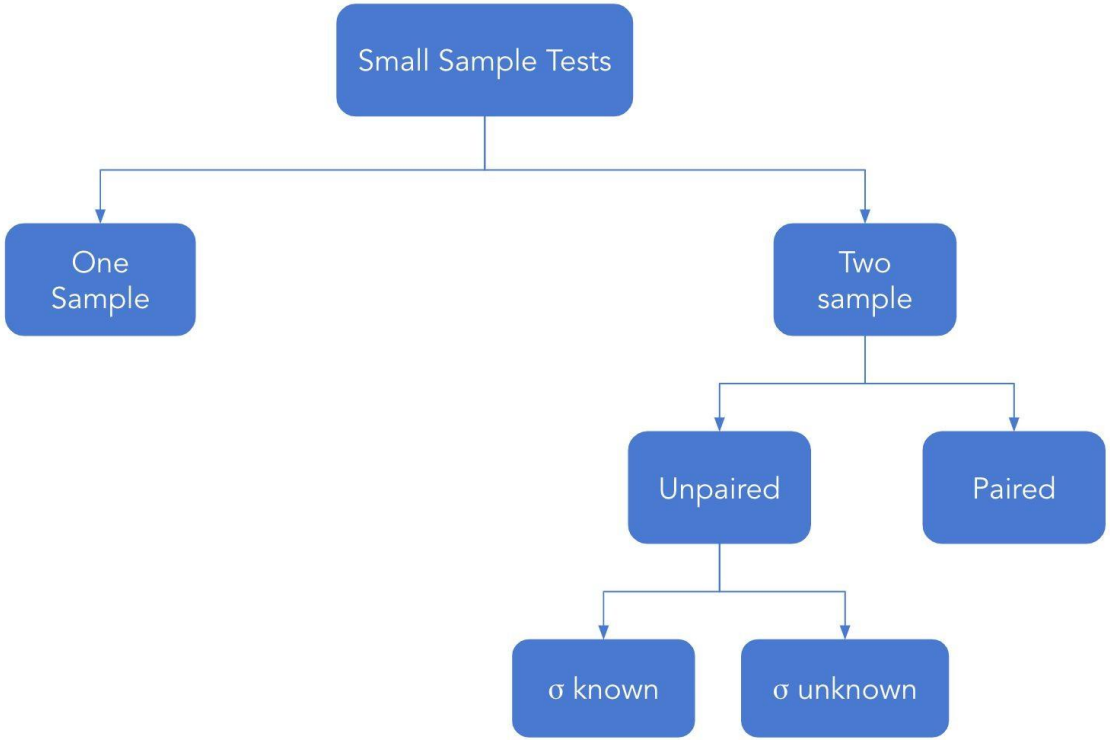


# Small Sample Tests

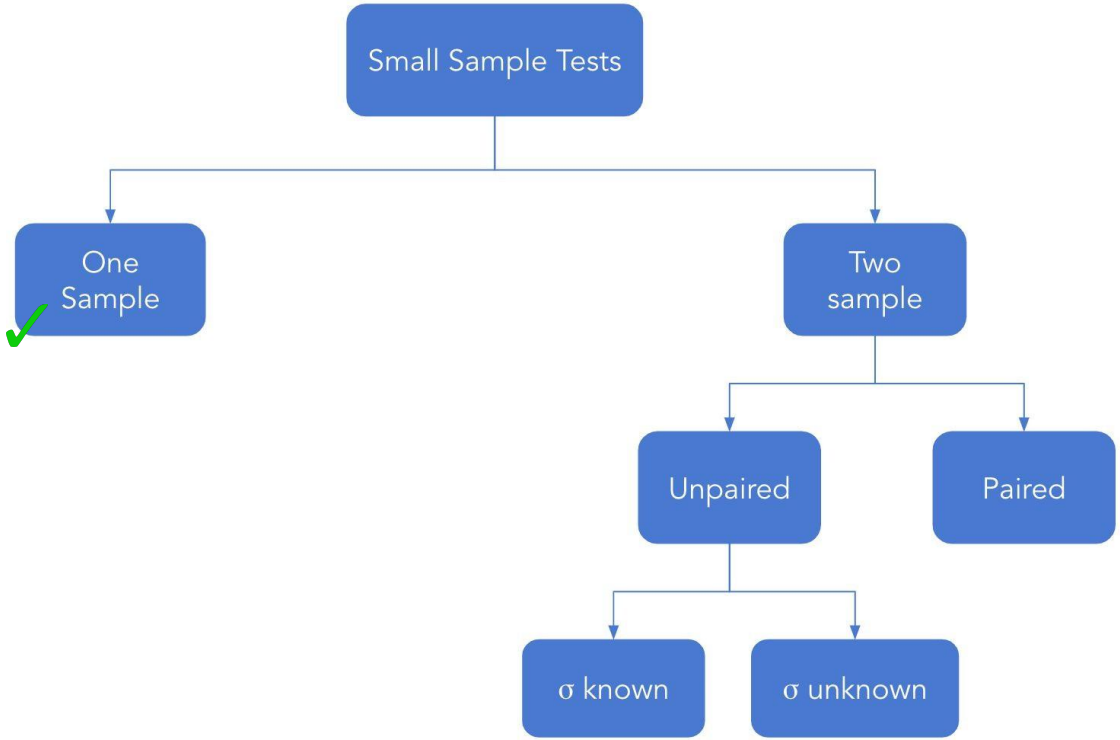
# Small sample tests

- For tests $n < 30$ are considered to be small sample tests
- Under the null hypothesis the test statistic follow exact distribution - t distribution, F distribution or $\chi^2$ distribution
- Small sample tests are regarded as exact tests

# Small sample tests



# Small sample tests

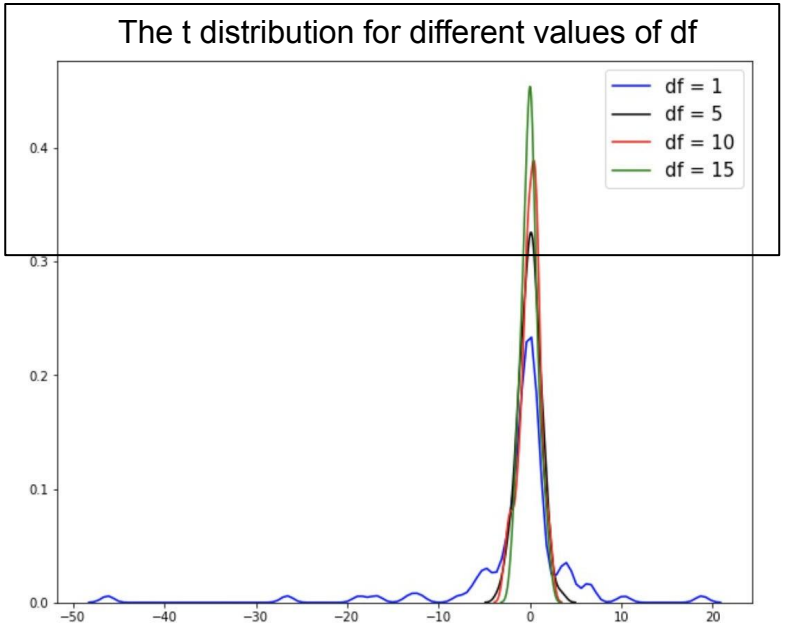


# One sample tests

- To test for population mean, if $\sigma^2$ is known the Z test is used
- If $\sigma^2$ is unknown, use the t-test
- The t-tests are based on the t-distribution
- The unknown $\sigma$ is replaced by sample standard deviation (s)

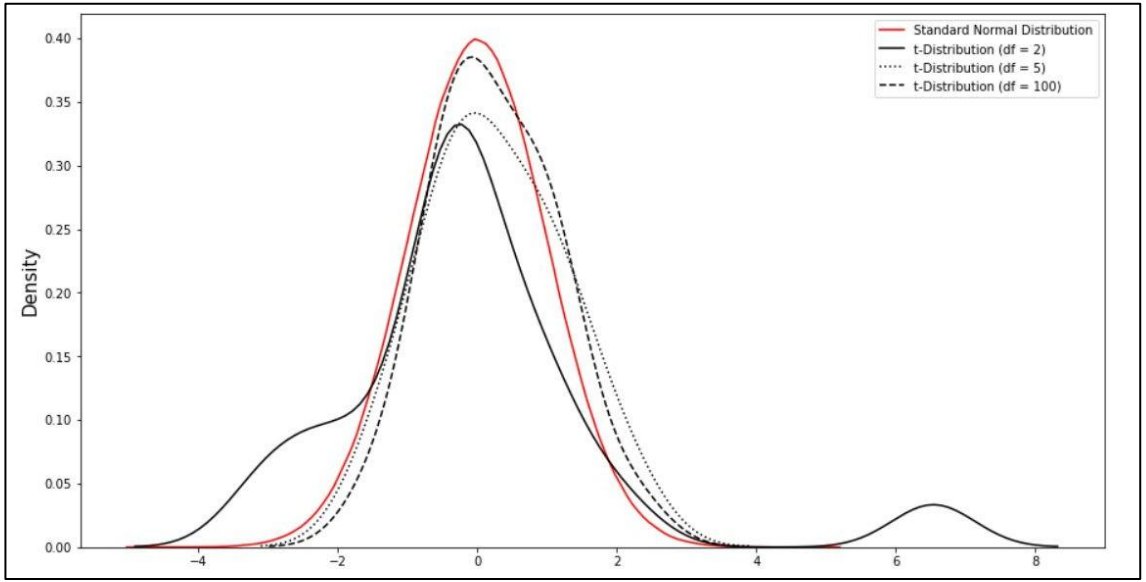
# The t-distribution

- It is based on the degrees of freedom
- The t distribution is a symmetric distribution about the mean
- Let *n* denote the degrees of freedom
- Its mean is 0 and variance is $\frac{n}{n-2}$ (for n>2)



# The t-distribution

- It is leptokurtic
- For large n it is mesokurtic



# Degrees of freedom

Degrees of freedom (d.f.) is the number of independent variables, i.e. they are free to vary.

Example:

Consider a system of equations:

- $Y = 8 + 5Z$
- $Z = X + 10$

Say if the value of Z is known, it is possible to find the value of other variables. Thus the d.f. is 1.

# One sample test - hypothesis

- The hypothesis to test whether the population mean is equal to a specified value when σ is unknown

$$H_0 : \mu = \mu_0 \quad \text{against} \quad H_1 : \mu \neq \mu_0$$

- It implies

$H_0$: The population mean is equal to $\mu_0$ (i.e $\mu = \mu_0$)

against $H_1$: The population mean is not equal to $\mu_0$ (i.e $\mu \neq \mu_0$)

# One sample test - hypothesis

- Like the one sample t- tests, it is possible to test for

$$H_0 : \mu \leq \mu_0 \quad \text{against} \quad H_1 : \mu > \mu_0$$

Or

$$H_0 : \mu \geq \mu_0 \quad \text{against} \quad H_1 : \mu < \mu_0$$

- Failing to reject $H_0$ implies that the null hypothesis is true

# One sample test - test statistic

- The test statistic t given by

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

$s^2$ is the sample estimate of population variance and is given by

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} \quad \text{or} \quad s^2 = \frac{n(\text{sample variance})}{n-1}$$

- Under $H_0$, the test statistic t follows t distribution with n-1 degrees of freedom

# Decision rule

| | $H_1$ | Based on critical region | Based on p-value | Based on confidence interval |
|---|---|---|---|---|
| For two tailed test | $\mu \neq \mu_0$ | Reject $H_0$ if $\lvert t \rvert \geq t_{df,\alpha/2}$ | Reject $H_0$ if p-value is less than or equal to the level of significance | Reject $H_0$ if $\mu_0$ does not lie in the confidence interval |
| For left tailed test | $\mu < \mu_0$ | Reject $H_0$ if $t \leq -t_{df,\alpha}$ | | |
| For right tailed test | $\mu > \mu_0$ | Reject $H_0$ if $t \geq t_{df,\alpha}$ | | |

# One sample t-test

Question:

A researcher is studying the growth of bacteria in waters of Lake Beach. The mean bacteria count of 100 per unit volume of water is within the safety level. The researcher collected 10 water samples of unit volume and found the mean bacteria count to be 94.8 with a sample variance of 72.66. Does the data indicate that the bacteria count is within the safety level? Test at the $\alpha = .05$ level. Assume that the measurements constitute a sample from a normal population.

# One sample t-test

Solution:

The mean bacteria count of 100 per unit volume of water is within the safety level.
i.e. $\mu = 100$

Let X: bacteria count per unit volume

The researcher collected 10 water samples of unit volume and found the mean bacteria count to be 94.8 with a sample variance of 72.66.
i.e. $n = 10$, $\bar{X} = 94.8$ and $s^2 = 72.66$

# One sample t-test

Solution:

To test: The bacteria count is within the safety level

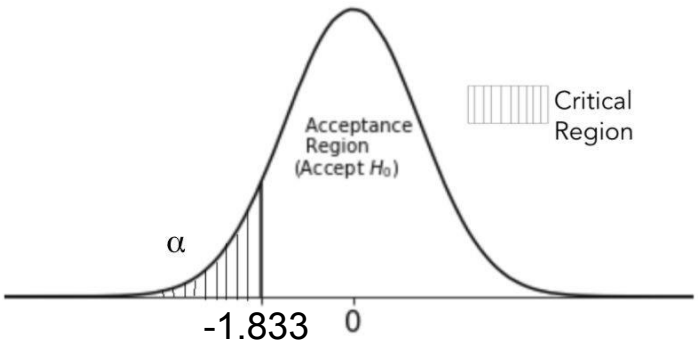
i.e. $H_0 : \mu \geq 100$ against $H_1 : \mu < 100$

The test statistic is

$$t = \frac{\bar{X}-\mu}{s/\sqrt{n}} = \frac{94.8-100}{\sqrt{72.66/10}} = -1.929$$

Here t follows $t_{9,0.05}$

From the table we have $t_{9,0.05} = -1.833$



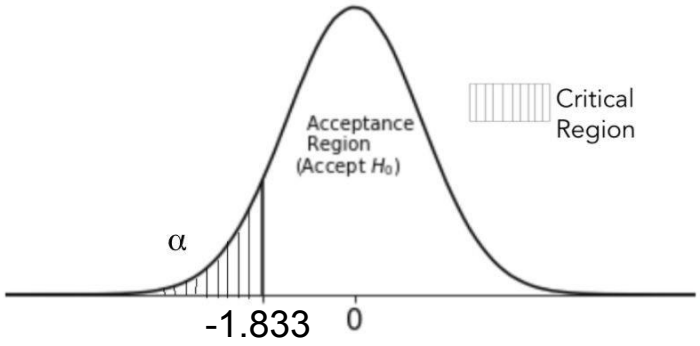
# One sample t-test

Solution:

$t_{critical}$ (-1.833) < $t_{calc}$ (-1.929)

Since the critical value is less than the calculated statistic value.

We reject the $H_0$, and can conclude that the average bacteria per unit volume (true mean) is within the safety levels.




As test statistic (=-1.929) > critical value (=-1.83), we reject $H_0$
As p-value < 0.05, we reject $H_0$.
.

# One sample t-test

Question:

A pharmaceutical company claims that a new drug reduces blood pressure by an average of 10 mmHg. A clinical trial is conducted on 25 patients, and the mean reduction in blood pressure is found to be 8.5 mmHg with a standard deviation of 2.2 mmHg. Test the company's claim at the $\alpha = 0.05$ level. Assume that the blood pressure reductions are normally distributed.

# Solution:

**Null Hypothesis ($H_0$):** $\mu = 10$ (The drug reduces blood pressure by 10 mmHg)

**Alternative Hypothesis ($H_1$):** $\mu \neq 10$ (The drug does not reduce blood pressure by 10 mmHg)

**Test Statistic:** $t = (\bar{x} - \mu) / (s/\sqrt{n}) = (8.5 - 10) / (2.2/\sqrt{25}) = -3.41$

**Degrees of Freedom (df):** $n - 1 = 25 - 1 = 24$

$\alpha = 0.05$ and $df = 24$ to be **±2.064.**

calculated t-value (-3.41) < critical t-value (-2.064), we reject the null hypothesis.

**Conclusion:** There is sufficient evidence to conclude that the drug does not reduce blood pressure by an average of 10 mmHg.

# One Sample - t test for Mean (σ unknown)

$$t_{STAT} = \frac{\bar{X} - \mu}{s/\sqrt{(n)}}$$

- where is the mean of the sample
  - μ is the mean of the population
  - s is the Sample standard deviation
  - n is the number of observations

- Here, t follows a t distribution having n-1 degrees of freedom

# One-Sample test - t test for Mean (σ unknown)

**Example**: Consider amount from Sales invoice of a nearby restaurant given the population mean is 120 USD.

108.98, 152.22, 111,45, 110.59, 127.46, 107.26, 93.32, 91.97, 111.56, 75.71, 128.58, 135.11

Test whether there is a significant deviation between your sample mean and the population mean?

**Solution:**

We observe from the sample,

Sample mean = 1354.21 / 12 = 112.85
Sample Std. deviation = 20.80
$n = 12$
$\mu = 120$
$s = 20.80$

# One-Sample test - t test for Mean (σ unknown)

**Step 1:** State the null and alternative hypothesis

**Ho: $\mu = 120$**

**Ha: $\mu \neq 120$**

**Step 2:** Choose the level of significance, α to be 5% and sample size is 12

**Step 3:** Choose the appropriate test statistic. Since σ is unknown, you use the t distribution and $\mathbf{t_{STAT}}$ as test statistic.

**Step 4:** Determine the critical value that divides the rejection and non-region regions.

For a given sample size n, the test statistic follows a t distribution with n-1 degrees of freedom. The alternative hypothesis has inequality sign and therefore has **two tails**. The area of rejection of the t distribution's left (lower) tail is 0.025, and the area in the rejection region of the t distribution's right (upper) tail is also 0.025. The t critical values are (-2.201 and 2.201) for 11 degrees of freedom at 5 % level of significance.

# One-Sample test - t test for Mean (σ unknown)

**Using Critical Value approach**

**Step 5:** Collect the data, organize the results and compute the value of the test

statistic. t = (112.85 - 120) / (20.80 / sqrt(12)) = -1.191

**Step 6:** State the statistical decision and the managerial conclusion.

- t is not in the region of rejection because $t_{STAT} < t_{CRITICAL}$ (-2.201) where $t_{STAT}$= - 1.191 Because the test statistic falls in the non-rejection region, the statistical decision is not to reject the null hypothesis.
- The managerial conclusion is that insufficient evidence exists to prove that mean amount per invoice has changed.
- No Corrective action on sales process is required.

# One-Sample test - t test for Mean (σ unknown)

**P value approach**

**Step 5**: Collect the data, organize the results and compute the p value of the test

statistic. t = (112.85 - 120) / (20.80 / sqrt(12)) = -1.1908

P(X > t = -1.1908, 11 df) = 0.259

**Step 6:** State the statistical decision and the managerial conclusion.

Since P-value (=0.259) > 0.05 (our level of significance), we do not reject null hypothesis. So statistical decision is not to reject the null hypothesis.

The managerial conclusion is that insufficient evidence exists to prove that mean amount per invoice has not changed.

**No Corrective action on sales process is required.**

Example: Salary of Management trainees with MBA from Premier institute and other institutes are given below:

Assume that the salary of Management trainees with MBA from Premier and non-Premier institutes follow normal distribution.
Check whether the difference in monthly salary is more than 5000 for Management trainees with MBA from Premier Institutes.

| Type of Management Institute | Sample Size | Mean monthly salary (INR) | Population standard deviation |
|---|---|---|---|
| Premier | 120 | 67500 | 7200 |
| Non-Premier | 45 | 58950 | 4600 |

# Two-Sample test - Z test for Mean (σ known)

**Solution**

We observe from the data,

n1 = 120 and n2 = 45 $\bar{x}_1 = 67500$ , $\bar{x}_2 = 58950$

$\sigma_1 = 7200$; $\sigma 2 = 4600$

**Step 1:** State the null and alternative hypothesis

$H_o$: $\mu 1 - \mu 2 \leq 5000$

$H_a$: $\mu 1 - \mu 2 > 5000$

**Step 2:** Choose the level of significance, α to be 5%.

**Step 3:** Choose the appropriate test statistic. Since σ is known, you use the Z distribution and Z as test statistic.

# Two-Sample test - Z test for Mean (σ known)

**Step 4:** Determine the critical value that divides the rejection and non-region regions.

The alternative hypothesis has > sign and therefore has one **tail**. The area of rejection of the Z distribution's right (upper) tail is 0.05. The Z critical value is 1.64 at 5 % level of significance.

**Using Critical Value approach**

**Step 5:** Collect the data, organize the results and compute the value of the test statistic.

$$z = \frac{((67500 - 58950) - 5000)}{\sqrt{\frac{7200^2}{120} + \frac{4600^2}{45}}} = 3.737$$

**Step 6:** State the statistical decision and the managerial conclusion.

- Z is in the region of rejection because $Z > Z_{CRITICAL}$ (= 1.64) where Z = 3.7374
- So, the statistical decision is to reject the null hypothesis.
- The managerial conclusion is that sufficient evidence exists to prove that difference in monthly salary is at least 5000 or more for Management trainees with MBA from Premier Institutes than others.

# Two-Sample test - Z test for Mean (σ known)

**Step 6 :** Determine the critical value that divides the rejection and non-region regions.

The alternative hypothesis has > sign and therefore has one **tail**. The area of rejection of the Z distribution right (upper) tail is 0.05. The Z critical value is 1.64 at 5 % level of significance.

**P Value approach**

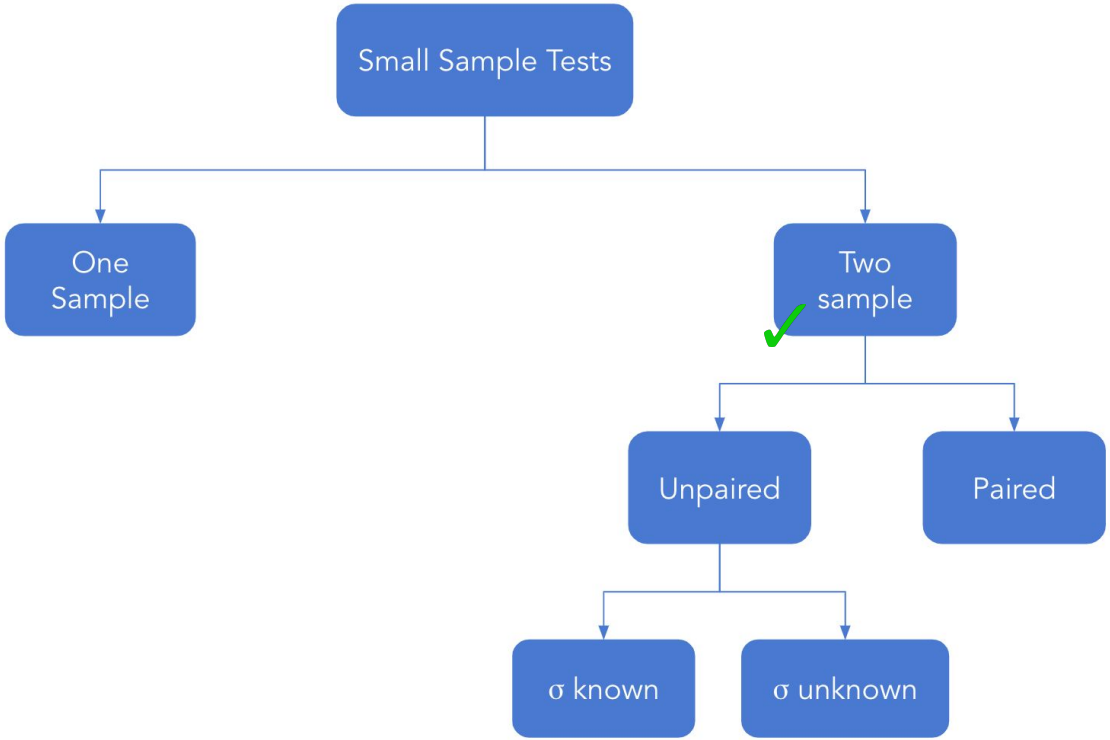
**Step 7:** Collect the data, organize the results and compute the value of the test statistic.

P-value = 9.296022e-05

**Step 8:** State the statistical decision and the managerial conclusion.

- Since P-value (= 9.296022e-05) < 0.05 (our level of significance), we reject null hypothesis.
- The managerial conclusion is that sufficient evidence exists to prove that difference in monthly salary is at least 5000 or more for Management trainees with MBA from Premier Institutes than others.

# Small sample tests



# t Test - 2 Sample

# Two sample tests

- Like the two sample Z test, the two sample t test are used to compare the equality of means of two populations for unpaired data

- For unpaired data, test relative performance two machineries, investment portfolios

# Two sample tests for unpaired data

- To test population means of two populations
- Let there be two different populations such that they follow normal distribution
- Two samples of sizes $n_1$ and $n_2$ drawn from normal population $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ respectively
- The two samples are independent of each other

# Two sample test - hypothesis

- The hypothesis to test whether the population means are equal

$H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_1$

- It implies

$H_0$: The two population means are equal (i.e $\mu_1 = \mu_2$)

against $H_1$: The two population means are not equal $\mu_0$ (i.e $\mu_1 \neq \mu_2$)

- The one sided hypothesis are

$H_0 : \mu_1 \leq \mu_2$ against $H_1 : \mu_1 > \mu_2$ Or $H_0 : \mu_1 \geq \mu_2$ against $H_1 : \mu_1 < \mu_2$

# Two sample tests - test statistic

- The test statistic is Z given by

$\sigma_1^2$ and $\sigma_2^2$ are known

$$Z = \frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}}$$

# Two sample tests - test statistic

- The test statistic is Z given by

$\sigma_1^2$ and $\sigma_2^2$ are known

$$Z = \frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}}$$

$\sigma_1^2$ and $\sigma_2^2$ are unknown

$$t = \frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{s\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$$

Where $s^2$ is the pooled sample variance given by $s^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}$

- If $\sigma_1^2$ and $\sigma_2^2$ are not known then they are replaced the sample estimates $s_1^2$ and $s_2^2$ respectively

# Two sample tests - test statistic

- If $\sigma_1^2$ and $\sigma_2^2$ are not known then they are replaced the sample estimates $s_1^2$ and $s_2^2$ respectively
- The pooled sample variance ($s^2$) is used
- Under $H_0$, the test statistic follows t distribution with $n_1 + n_2 - 2$ df
- Failing to reject $H_0$ implies that the two population means are equal (i.e $\mu_1 = \mu_2$)

# Two sample test for unpaired data

Question:

An experiment was conducted to compare the pain relieving hours of two new medicines. Two groups of 14 and 15 patients were selected and were given comparable doses. Group 1 was given medicine 1 and group 2 was given medicine 2. Following data is obtained from the two samples. Test whether the two populations give the same mean hours of relief. Assume the data comes from normal distribution has equal variance. [Use $\alpha = 0.01$]

| | Medicine 1 | Medicine 2 |
|---|---|---|
| Mean of hours of relief | 6.4 | 7.3 |
| S.D of hours of relief | 1.4 | 1.5 |

# Two sample test for unpaired data

Solution:

Let X: patients receiving medicine 1
Y: patients receiving medicine 2

To test $H_0$: $\mu_1 = \mu_2$ against $H_1$: $\mu_1 \neq \mu_2$

The pooled variance is

$$s^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2} = \frac{(13)1.4^2+(14)1.5^2}{14+15-2} = 2.11$$

# Two sample test for unpaired data

Solution:

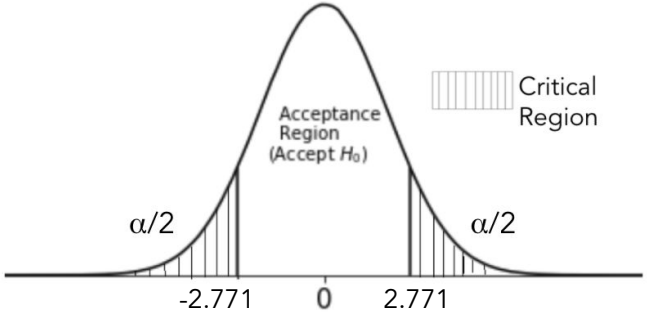
The test statistic is

$$t = \frac{(\bar{X}-\bar{Y})}{s\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{(6.4-7.3)}{1.4525\sqrt{\frac{1}{14}+\frac{1}{15}}} = -1.667$$

Decision rule: Reject $H_0$ if $|t_{cal}| \geq t_{n1+n2-2,\ \alpha/2}$

$t_{n1+n2-2,\ \alpha/2} = t_{27,0.005} = 2.771$

Since 2.771>|-1.667|, we fail to reject $H_0$.




The two medicines have the same mean hours of sleep.
As test statistic (=-1.66) < critical value (=-2.59), we fail to reject $H_0$.
As p-value > 0.01, we fail to reject $H_0$.

A company wants to compare the average lifespan of two different types of batteries, A and B. They know from past experience that the population standard deviations for both types of batteries are 20 hours.A random sample of 40 batteries of type A has a mean lifespan of 300 hours. A random sample of 35 batteries of type B has a mean lifespan of 285 hours.At the 0.05 significance level, is there a difference in the mean lifespan of the two types of batteries?

**Null Hypothesis ($H_0$):** $\mu_1 = \mu_2$ (There is no difference in the mean lifespans of the two battery types.)
**Alternative Hypothesis ($H_1$):** $\mu_1 \neq \mu_2$ (There is a difference in the mean lifespans of the two battery types.)

$\alpha = 0.05$

$z = (\bar{x}_1 - \bar{x}_2) / \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$

$z = (300 - 285) / \sqrt{(20^2/40 + 20^2/35)} \approx 2.04$

Since the calculated z-value (2.04) is greater than the critical value (1.96), we reject the null hypothesis.

**Conclusion:**
There is sufficient evidence to conclude that there is a significant difference in the mean lifespans of the two types of batteries

A researcher wants to compare the mean annual salaries of employees at two different companies, Company A and Company B. A random sample of 30 employees from Company A has a mean salary of $55,000 with a standard deviation of $8,000. A random sample of 40 employees from Company B has a mean salary of $52,000 with a standard deviation of $7,000.

At the 0.01 significance level, is there a difference in the mean annual salaries of the two companies?

**Null Hypothesis ($H_0$):** $\mu_1 = \mu_2$ (There is no difference in the mean annual salaries of the two companies.)
**Alternative Hypothesis ($H_1$):** $\mu_1 \neq \mu_2$ (There is a difference in the mean annual salaries of the two companies.)

$\alpha = 0.01$

$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{((s_1^2/n_1) + (s_2^2/n_2))}$

$t = (55000 - 52000) / \sqrt{((8000^2/30) + (7000^2/40))} \approx 2.26$

$df \approx ((s_1^2/n_1 + s_2^2/n_2)^2 / ((s_1^2/n_1)^2/(n_1-1) + (s_2^2/n_2)^2/(n_2-1)))$

Using a t-distribution table or a statistical software, we find the critical values for a two-tailed test at $\alpha = 0.01$ and df $\approx 66$ to be approximately $\pm 2.65$.

Since the calculated t-value (2.26) is between the critical values (-2.65 and 2.65), we fail to reject the null hypothesis.

# Independent Sample t Test

An aptitude test was conducted in which groups of engineering and accounting executives participated. Is there a significant difference in the performance the groups?

| | Scores | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Engineers | 125 | 115 | 119 | 85 | 97 | 107 | 125 | 125 | 118 |
| Accountants | 112 | 98 | 109 | 96 | 77 | 70 | 114 | 100 | |

| | Formula | Engineers | Accountants |
|---|---|---|---|
| Mean | | 112.889 | 97 |
| Variance | | 196.611 | 256.857 |
| Observations | | 9 | 8 |

# Independent Sample t Test

## Solution:

| t-Test: Two-Sample Assuming Equal Variances | | |
|---|---|---|
| | Engineers | Accountants |
| Mean | 112.889 | 97 |
| Variance | 196.611 | 256.857 |
| Observations | 9 | 8 |
| Pooled Variance | 224.726 | |
| Hypothesized Mean Difference | 0 | |
| df | 15 | |
| t Stat | 2.181 | |
| P(T<=t) one-tail | 0.023 | |
| t Critical one-tail | 1.753 | |
| P(T<=t) two-tail | 0.045 | |
| t Critical two-tail | 2.131 | |

**This is a 2 Tailed Test. Since positive t > $t_{critical}$, we reject null hypothesis. There is significant difference is performance.**

- The test is carried out under the assumption that the samples are drawn from independent normal population

- If the populations from which the samples are drawn do not have equal variance then the t-test can not be use. In this case the Welch test is used

# Matched / Paired Sample Hypothesis Test

- Compare a population parameter such as mean before and after intervention.
- In a paired t test, the data related to a parameter is captured twice, once before the intervention and once after the intervention. (Example: Amount of alcohol consumed by people before and after marriage break-up)

*Assumptions:*

The differences between the estimated parameter before and after intervention follow a normal distribution.

- Test statistic, $t = (D - \mu_D) \quad / (S_D / \sqrt{(n)})$ follows a t distribution with n -1 degrees of freedom.
- Here D is the mean difference in the estimated parameter values before and after the treatment
- $\mu_D$ is the hypothesized mean difference and $S_D$ is the corresponding standard deviation.

# Two sample tests for paired data

- For paired data effectiveness of some training/treatment is measured
- The observations are recorded on the same individual/item twice resulting in pairs of observations

Example:

A energy drink manufacturing company wants to test if sales increase after they advertise the drink with a life sized picture of a well-known athlete.

# Two sample tests for paired data

- Let $\{(X_i, Y_i), i = 1, 2, 3, \ldots, n\}$ where X and Y are paired data
- Let $\mu_X$, $\mu_Y$ be the mean of data from X and Y respectively
- Define $d_i = y_i - x_i$
- Let $\mu_d = \mu_y - \mu_x$
- In paired t-test, we test for

$$H_0 : \mu_d = \mu_0 \quad \text{against} \quad H_1 : \mu_d \neq \mu_0$$

$$H_0 : \mu_d \leq \mu_0 \quad \text{against} \quad H_1 : \mu_d > \mu_0$$

$$H_0 : \mu_d \geq \mu_0 \quad \text{against} \quad H_1 : \mu_d < \mu_0$$

# Two sample tests for paired data

- Suppose mean of $d_i$ = $\bar{d} = \frac{\sum d_i}{n}$ and variance of $d_i$ = $s^2 = \frac{\sum (d_i - \bar{d})^2}{n-1}$
- The test statistic is

$$t = \frac{\bar{d} - \mu_d}{s/\sqrt{n}}$$

- Under $H_0$, the test statistic follows t-distribution with n-1 degrees of freedom
- Failing to reject $H_0$, implies that the null hypothesis is true

# Two sample t-test for paired data

Question:

An energy drink distributor claims that a new advertisement poster, featuring a life-size picture of a well-known athlete. For a random sample of 10 outlets, the following data was collected. Test that the null hypothesis that there the advertisement was effective in increasing the sales

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Before | 33 | 32 | 38 | 45 | 37 | 47 | 48 | 41 | 45 |
| After | 42 | 35 | 31 | 41 | 37 | 36 | 49 | 49 | 48 |

Test the hypothesis using critical region technique. [Use $\alpha = 0.05$].

# Two sample t-test for paired data

Solution:

To test,

$H_0$: The advertisement was not effective against $H_1$: The advertisement was effective

$(\mu_d \leq 0)$ $(\mu_d > 0)$

Compute $d_i$

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Before | 33 | 32 | 38 | 45 | 37 | 47 | 48 | 41 | 45 |
| After | 42 | 35 | 31 | 41 | 37 | 36 | 49 | 49 | 48 |
| $d_i=y_i-x_i$ | 9 | 3 | -7 | -4 | 0 | -11 | 1 | 8 | 3 |

# Two sample t-test for paired data

Solution:

To test,

$H_0$: The advertisement was not effective ( $\mu_d \leq 0$) against $H_1$: The advertisement was effective ($\mu_d > 0$)

We have mean $\bar{d} = \frac{\sum d_i}{n} = 0.22$ and its variance $s^2 = \frac{\sum (d_i - \bar{d})^2}{n-1} = 43.6944$

The test statistic is

$$t = \frac{d - \mu_d}{s/\sqrt{n}} = \frac{0.22 - 0}{6.6101/\sqrt{9}} = 0.1008$$

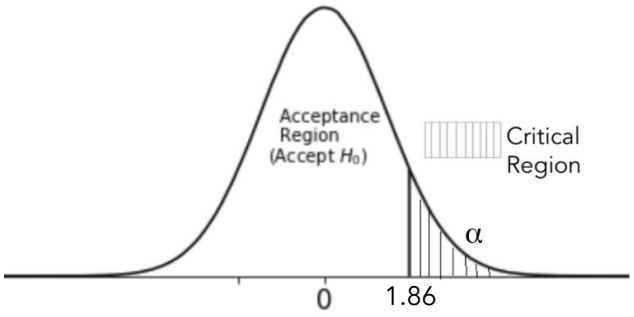
# Two sample t-test for paired data

Solution:

Decision Rule: Reject $H_0$, if $t_{calc} \geq t_{df,\alpha}$

Here $t_{df,\alpha} = t_{n-1,\alpha} = t_{8,0.05} = 1.86$

Since 1.86 > 0.0998, fail to reject $H_0$




Thus, there is no effect of the advertisement.

The test statistic (=0.1008) < critical value (=1.86), also the p-value > 0.05, thus we fail to reject $H_0$.

A researcher believes that Cholesterol level of participants of a meditation program will be lesser after completing the program than before attending the program. Check the claim at 5% level of significance.

Sample size = 50, D = 14 $S_D$ = 8.5 $\mu_D$ = 10 mg/dL

# Example of Paired Sample t Test

A researcher believes that Cholesterol level of participants of a meditation program will be lesser after completing the program than before attending the program. Check the claim at 5% level of significance.

Sample size = 50, D = 14 $S_D$ = 8.5 $\mu_D$ = 10 mg/dL

## Solution:

**Step 1**: State the null and alternative hypothesis

Ho: $\mu_D \leq 10$
Ha: $\mu_D > 10$

**Step 2**: Choose the level of significance, $\alpha$ to be 5% and sample size is 50

**Step 3**: Choose the appropriate test statistic. Since $\sigma$ is unknown, you use the t distribution and t as test statistic.

# Paired Sample t Test

**Using P Value approach**

**Step 4**: Determine the critical value that divides the rejection and non-rejection regions.

n=50, D = 14 $S_D$ = 8.5 $\mu_D$ = 10 mg/dL

$$t = \frac{(D - \mu_D)}{\left({s_D}/{\sqrt{n}}\right)} = \frac{(14 - 10)}{{8.5}/{\sqrt{50}}} = 3.328$$

Critical value for t when $\alpha = 0.05$ and degrees of freedom = n – 1 = 49 is 1.677

t = 3.328 with p value is .0008

# Paired Two-Sample t test

**Step 5**: State the statistical decision and the managerial conclusion.

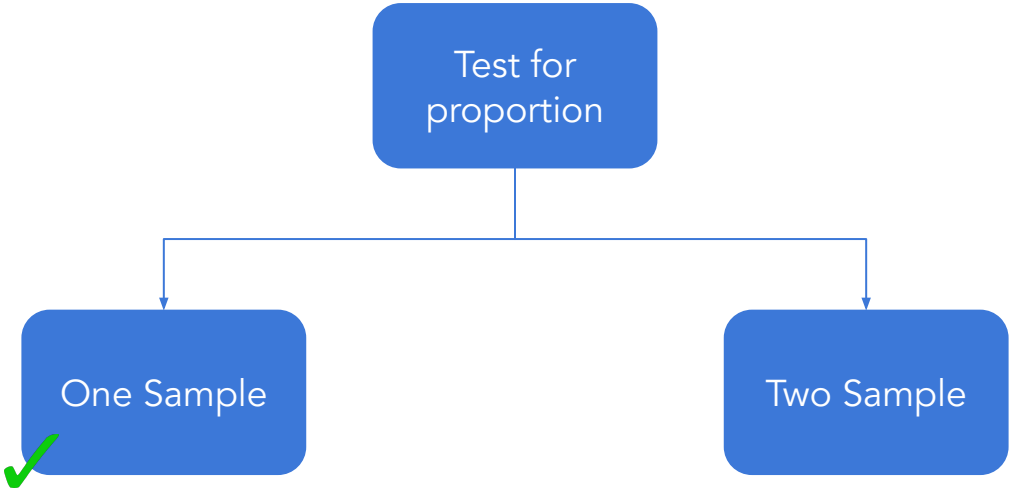
- t is in the region of rejection because p value (= 0.0008) < 0.05
- So, the statistical decision is to reject the null hypothesis.
- The managerial decision is that sufficient evidence exists to prove that cholesterol level of participants of meditation program reduces at least by 10 mg/dL after attending the meditation program at 5% level of significance.

# Test for Population Proportion

# Test for proportion

- For qualitative data the proportion of a desired characteristic is obtained
- Test for proportion:
  - One sample: Testing population proportion (P) is equal to a specified value ($P_0$)
  - Two sample: Testing equality of Two population proportions ($P_1 = P_2$)
- Similar to the tests of population mean

# Test for proportion



# One sample test - hypothesis

- The hypothesis to test the population proportion is equal to a specified value

$$H_0 : P = P_0 \quad \text{against} \quad H_1 : P \neq P_0$$

- It implies

$H_0$: The population proportion is equal to $P_0$

against $H_1$: The population proportion is not equal to $P_0$

- Failing to reject $H_0$ implies that the population proportion is equal to $P_0$

# Test for proportions

- The test statistic is given by

Sample proportion

Specified proportion

$$Z = \frac{p - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}}$$

Sample size

- Under $H_0$, the test statistic follows standard normal distribution

# One sample test for proportion - decision rule

| | $H_1$ | Based on critical region | Based on p-value | Based on confidence interval |
|---|---|---|---|---|
| For two tailed test | $P \neq P_0$ | Reject $H_0$ if $\lvert Z \rvert \geq Z_{\alpha/2}$ | Reject $H_0$ if p-value is less than or equal to the level of significance | Reject $H_0$ if $P_0$ does not lie in the confidence interval |
| For left tailed test | $P < P_0$ | Reject $H_0$ if $Z \leq -Z_\alpha$ | | |
| For right tailed test | $P > P_0$ | Reject $H_0$ if $Z \geq Z_\alpha$ | | |

# One sample test for proportion

Question:

From a sample 361 business owners had gone into bankruptcy due to recession. On taking a survey, it was found that 105 of them had not consulted any professional for managing their finance before opening the business. Test the null hypothesis that at most 25% of all businesses had not consulted before opening the business.

Test the claim using p-value technique. [Use $\alpha = 0.05$].

# One sample test for proportion

Solution:

From a sample 361 business owners had gone into bankruptcy due to recession.
i.e. $n = 361$

On taking a survey, it was found that 105 of them had not consulted any professional for managing their finance before opening the business.

Let X: business which did not consult before
$x = 105$

The sample proportion $(p) = x/n = 105/361 = 0.2909$

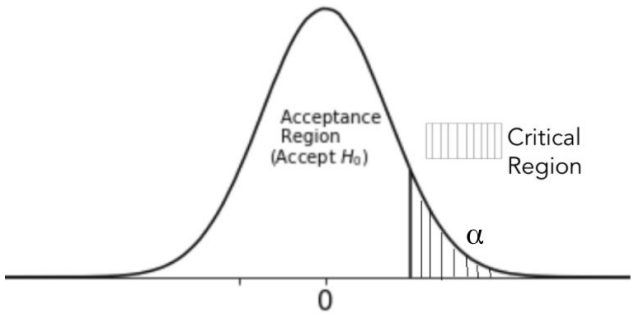
# One sample test for proportion

Solution:

To test: The null hypothesis that at most 25% of all businesses had not consulted before opening the business

Here $P_0 = 0.25$

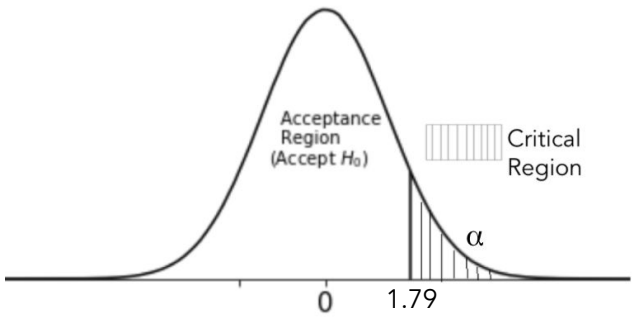
To test, $H_0$: $P \leq 0.25$ against $H_1$: $P > 0.25$



# One sample test for proportion

Solution:

The test statistic $Z = \frac{p - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}} = \frac{0.2909 - 0.25}{\sqrt{\frac{0.25(0.75)}{316}}} = 1.79$




The p-value = P(Z > 1.79) = 0.0367

As p-value < 0.05, reject $H_0$.

We may conclude that at least 25% of all businesses had not consulted before starting the business.

# Test for proportion

When the outcome is discrete or dichotomous (success or failure), the test objective is :

Compare the proportion of successes (P-hat or P) in a single population to a known proportion ($P_0$)

This test will assess whether or not a sample from a population represents the true proportion from the entire population.

The proportion P-hat is calculated using the ratio of no. of successes to the sample size (n)

$$\text{P-hat} = x/n$$

# Standard Deviation and Z score for Test of proportion

- Standard deviation (σ) of the sampling distribution

$$se = sqrt[ P_0 * ( 1 - P_0 ) / n ]$$

where $P_0$ is the hypothesized value of population proportion in the null hypothesis, and n is the sample size.

- Test statistic. The test statistic is a z-score (z) defined by the following equation.

$$z = (p - P_0) / se$$

where $P_0$ is the hypothesized value of population proportion in the null hypothesis, P is the sample proportion (sometimes referred to as P-hat), and σ is the standard deviation of the sampling distribution.

# Test of proportion with One Sample - Critical Value Approach

1. State the null hypothesis $H_o$ and the alternative hypothesis $H_A$.
2. Calculate the test statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

where $p_0$ is the null hypothesized proportion i.e., when $H_o$: $P=P_0$

P-hat is the sample proportion

3. Determine the critical region.
4. Make a decision. Determine if the test statistic falls in the critical region. If it does, reject the null hypothesis. If it does not, do not reject the null hypothesis.

# Test of proportion with One Sample - p-Value Approach

1. State the null hypothesis $H_o$ and the alternative hypothesis $H_A$.
2. Calculate the test statistic:

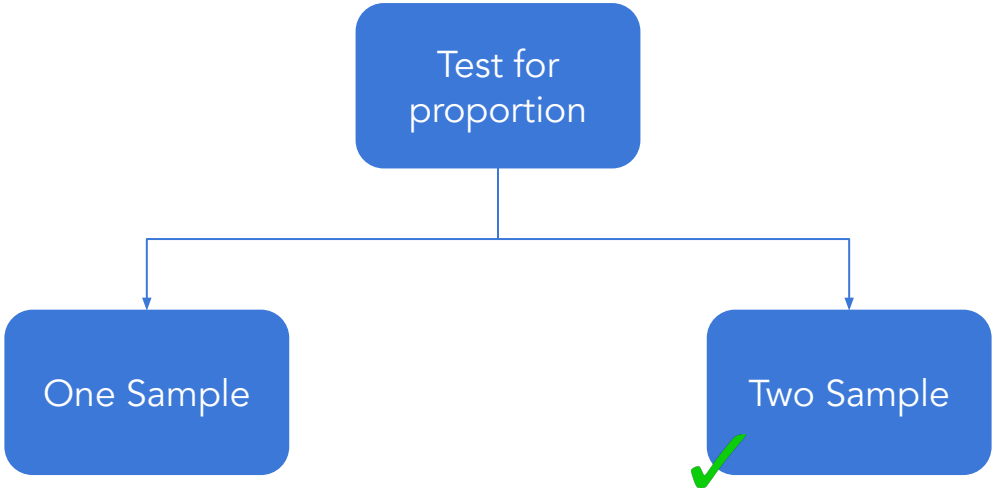
$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

where p0 is the null hypothesized proportion i.e., when $H_o$: $P=P_0$

P-hat is the sample proportion

3. Calculate the p-value.
4. Make a decision. Check whether to reject the null hypothesis by comparing p-value to α. If the p-value < α, then reject Ho; otherwise do not reject Ho.

# Test for proportion



# Test of proportion with Two Independent Samples

Objective:

Compare proportions of successes between the two groups. The relevant sample data are the sample sizes in each comparison group ($n_1$ and $n_2$) and the sample proportions ($P_1$-hat and $P_2$-hat) which are computed by taking the ratios of the numbers of successes ($x_1$ and $x_2$) to the sample sizes in each group, i.e.,

$$\hat{p}_2 = \frac{x_2}{n_2} \quad \text{and} \quad \hat{p}_1 = \frac{x_1}{n_1}$$

Test Statistic : $$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$ where $\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$

# Two sample tests for population proportion

- Let there be two samples sizes $n_1$ and $n_2$ from different populations of such that $x_1$ and $x_2$ are the number of specific items in each of them respectively

- Suppose these samples have proportions of specific items $p_1$ and $p_2$ respectively

- To test the equality of population proportion from which these samples are chosen

# Two sample test - hypothesis

- The hypothesis to test the population proportion

$$H_0 : P_1 = P_2 \quad \text{against} \quad H_1 : P_1 \neq P_2$$

- It implies

$H_0$: The two population proportions are equal ($P_1 = P_2$)

against $H_1$: The two population proportions are not equal ($P_1 \neq P_2$)

- Failing to reject $H_0$ implies that the two population proportions are equal

# Test for proportions

- The test statistic is given by

$$Z = \frac{p_1 - p_2}{\sqrt{\bar{P}(1-\bar{P})(\frac{1}{n_1} + \frac{1}{n_2})}}$$

where $\bar{P}$ is the proportion of pooled sample such that

$$\bar{P} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

- Under $H_0$, the test statistic follows standard normal distribution

# Two sample test for proportion - decision rule

| | $H_1$ | Based on critical region | Based on p-value | Based on confidence interval |
|---|---|---|---|---|
| For two tailed test | $P_1 \neq P_2$ | Reject $H_0$ if $\lvert Z \rvert \geq Z_{\alpha/2}$ | Reject $H_0$ if p-value is less than or equal to the level of significance | Reject $H_0$ if $P_1 - P_2$ does not lie in the confidence interval |
| For left tailed test | $P_1 < P_2$ | Reject $H_0$ if $Z \leq -Z_\alpha$ | | |
| For right tailed test | $P_1 > P_2$ | Reject $H_0$ if $Z \geq Z_\alpha$ | | |

# Two sample test for proportion

Question:

Steve owns a kiosk where sells two magazines - A and B in a month. He buys 100 copies of magazine A out of which 78 were sold and 70 copies of magazine B out of which 65 were sold. Is there enough evidence to say that magazine is B is more popular?

Test the claim using p-value technique. [Use $\alpha = 0.05$].

# Two sample test for proportion

Solution:

Steve owns a kiosk where sells two magazines - A and B in a month.

Let X: the number of magazines sold

Out of 100 copies of magazine A 78 are sold
Here, $x_1 = 78$ and $n_1 = 100$
Let $p_1$ be the proportion of sell of magazine A
$p_1 = x_1/n_1 = 78/100 = 0.78$

Out of 70 copies of magazine B 65 are sold
Here, $x_2 = 65$ and $n_1 = 70$
Let $p_2$ be the proportion of sell of magazine B
$p_2 = x_2/n_2 = 65/70 = 0.928$

# Two sample test for proportion

Solution:

To test, whether magazine B is more popular, i.e

$H_0: P_1 \geq P_2$ against $H_1: P_1 < P_2$

Where $P_1$: denotes population proportion of magazine A sold
$P_2$: denotes population proportion of magazine B sold

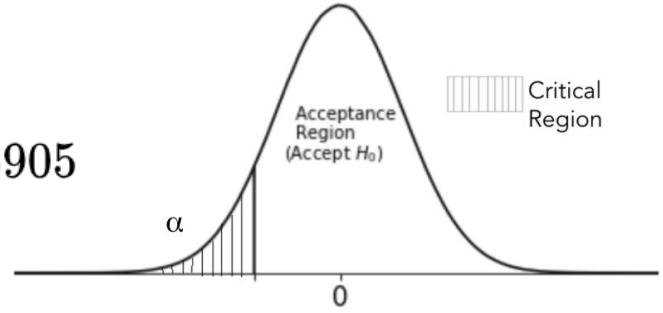
# Two sample test for proportion

Solution:

The pooled proportion is $\bar{P} = \frac{x_1+x_2}{n_1+n_2} = \frac{78+65}{100+70} = 0.84$

The test statistic is

$$Z = \frac{p_1 - p_2}{\sqrt{\bar{P}(1-\bar{P})(\frac{1}{n_1}+\frac{1}{n_2})}} = \frac{0.78-0.928}{\sqrt{0.84(1-0.84)(\frac{1}{100}+\frac{1}{70})}} = -2.5905$$



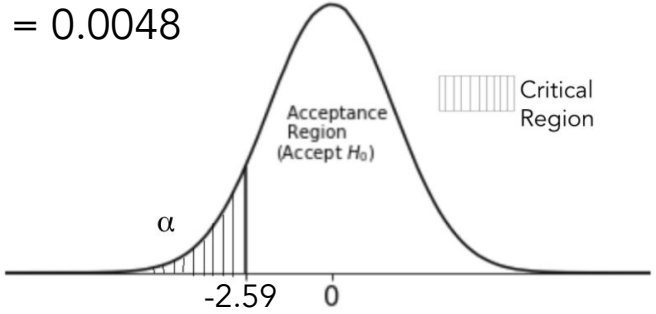
# Two sample test for proportion

Solution:

The test statistic Z = -2.5905

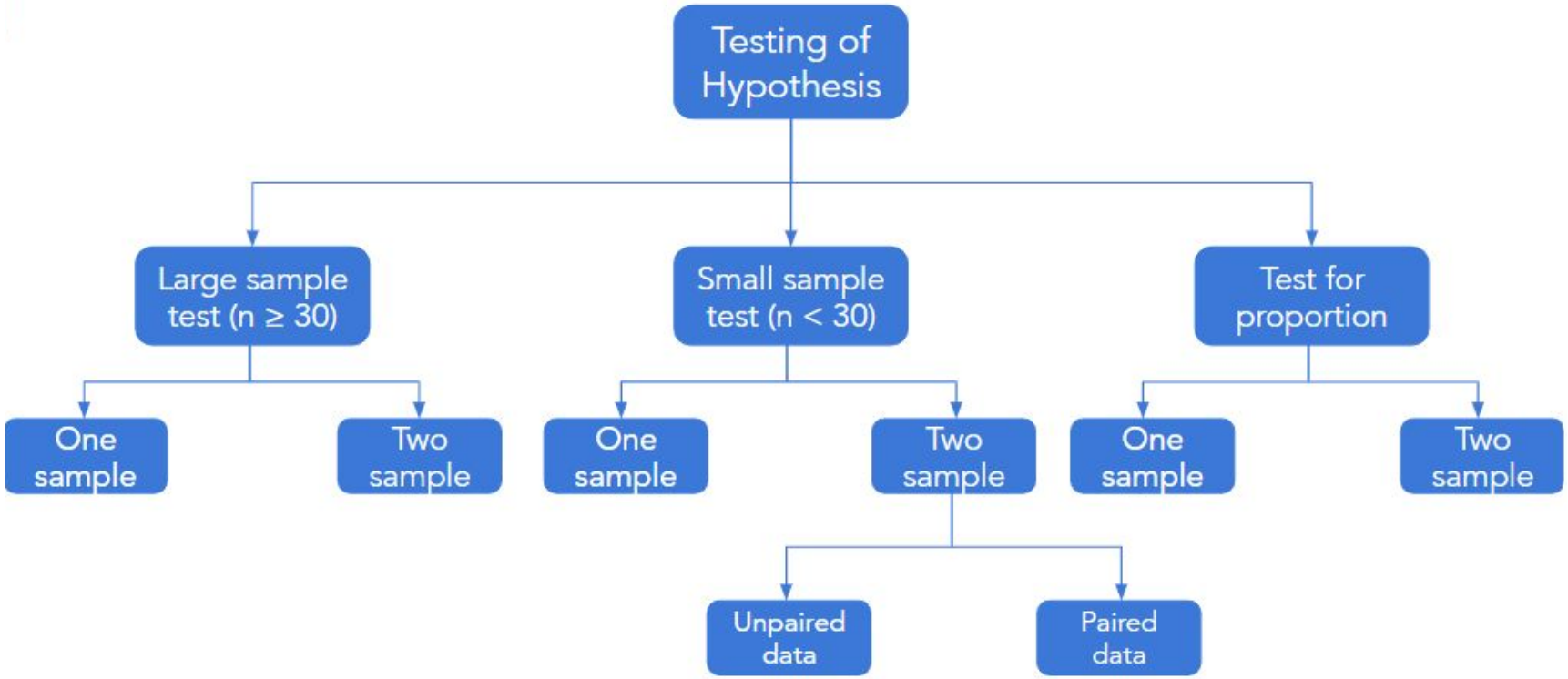
The p-value = P(Z < $Z_{calc}$, under $H_0$) = P(Z < -2.5905, μ = 13) = 0.0048

Since p-value < 0.05, we reject $H_0$.




Thus there is enough evidence to conclude that magazine is B is more popular.

# Summary


Testing of Hypothesis
Large sample test (n ≥ 30)
Small sample test (n < 30)
Test for proportion
One sample
Two sample
One sample
Two sample
One sample
Two sample
Unpaired data
Paired data

# Parametric & Non parametric tests

- The tests considered so far have two features:
  - The probability distribution of the samples was assumed to be known
  - The hypothesis test was about the parameter of the probability distribution
- These tests are known as the parametric tests
- The times when these assumptions are not satisfied, use the non-parametric tests

Thank You